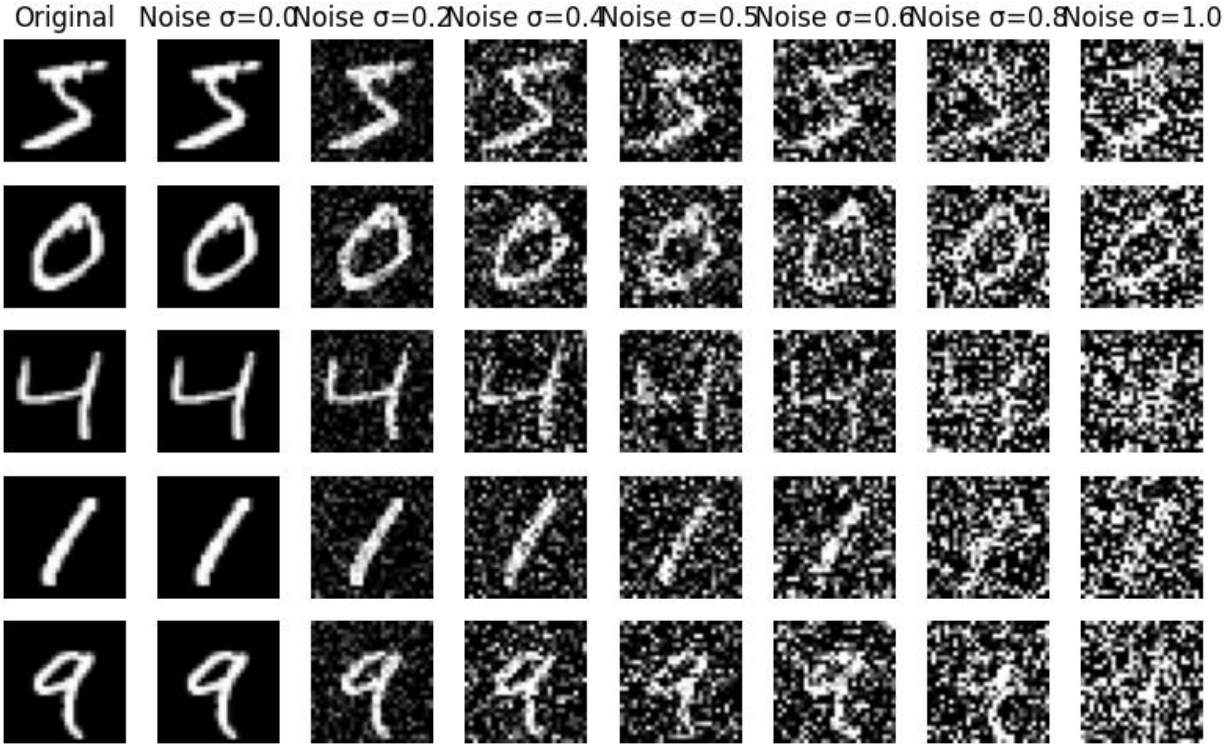
Assignment 5: Diffusion Models

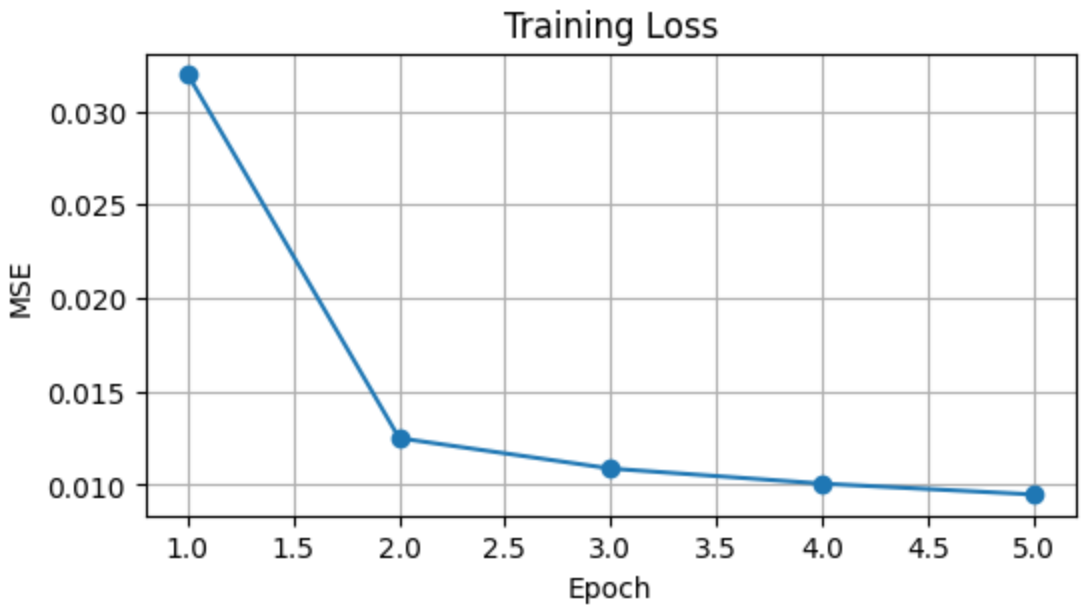
Name(s): Yanjin Li
NetID(s): yanjin2

Part 1: Single-Step Denoising UNet

1. A visualization of the noising process using σ = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0] (Figure 3)

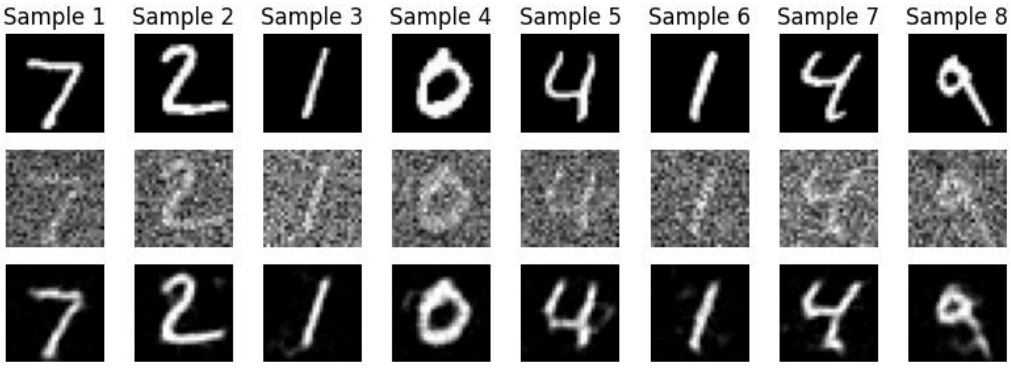


2. A training loss curve plot every few iterations during the whole training process (Figure 4)

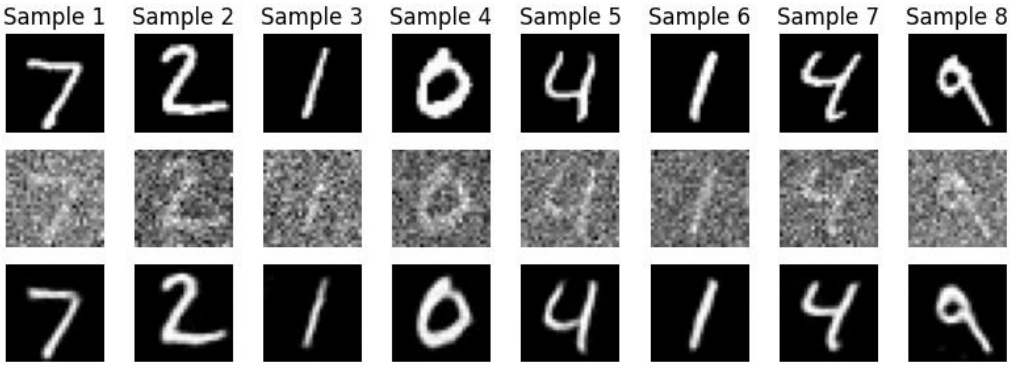


3. Sample results on the test set after the first and the 5th epoch. (Figures 5 and 6)

1st:

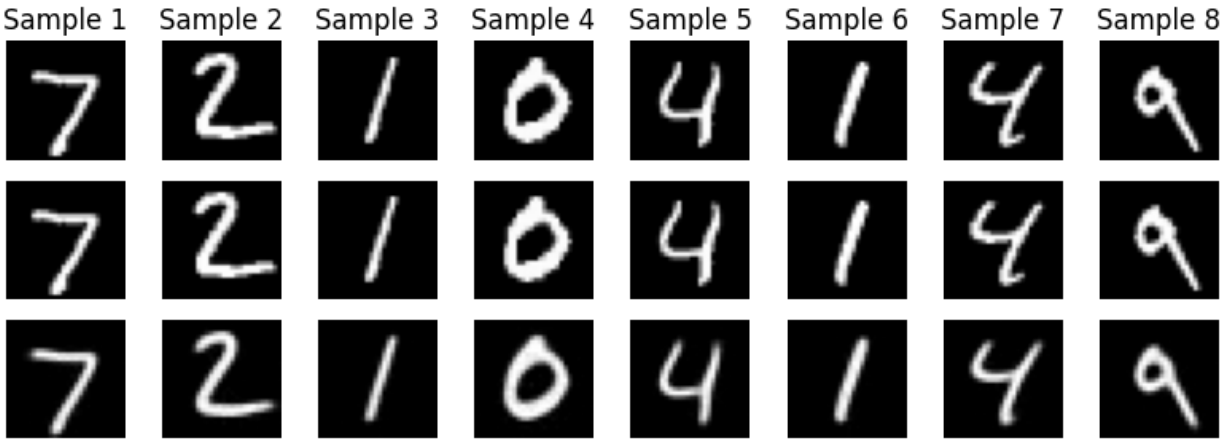


5th:

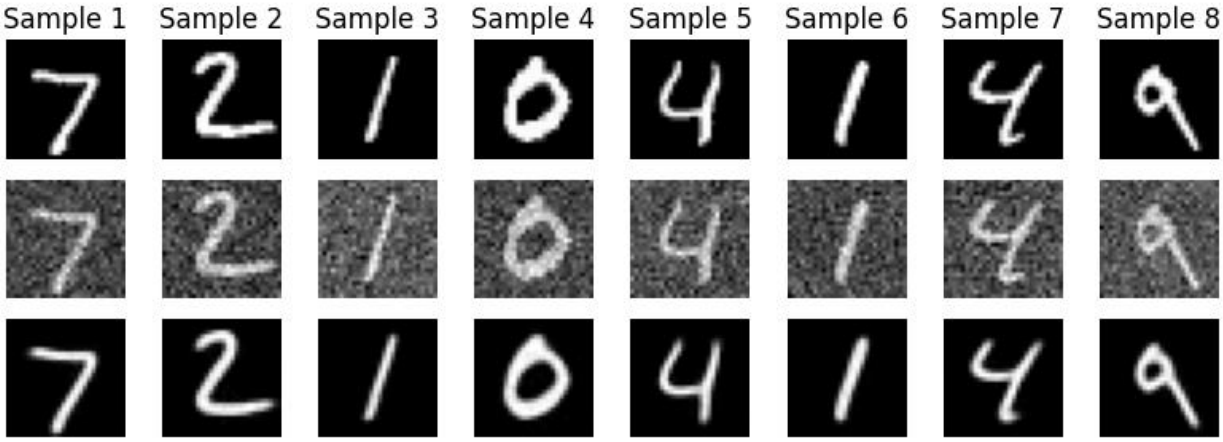


4. Sample results on the test set with out-of-distribution noise levels after the model is trained. Keep the same image and vary σ = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0] (Figure 7)

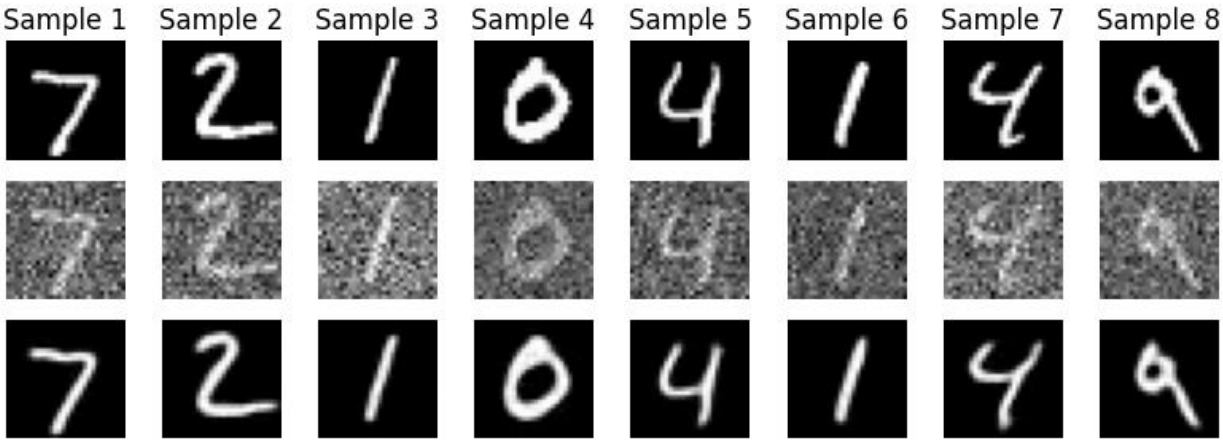
σ = 0.0



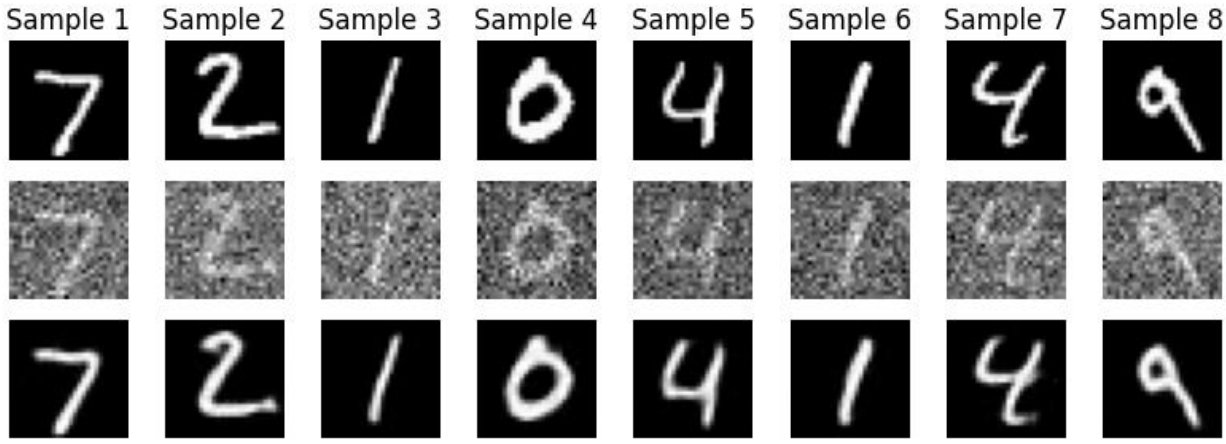
σ = 0.2



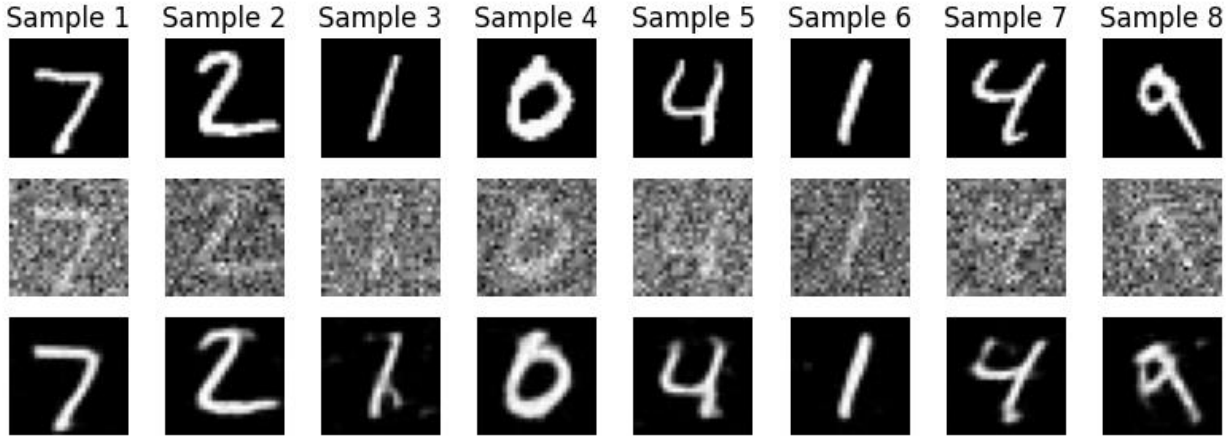
σ = 0.4



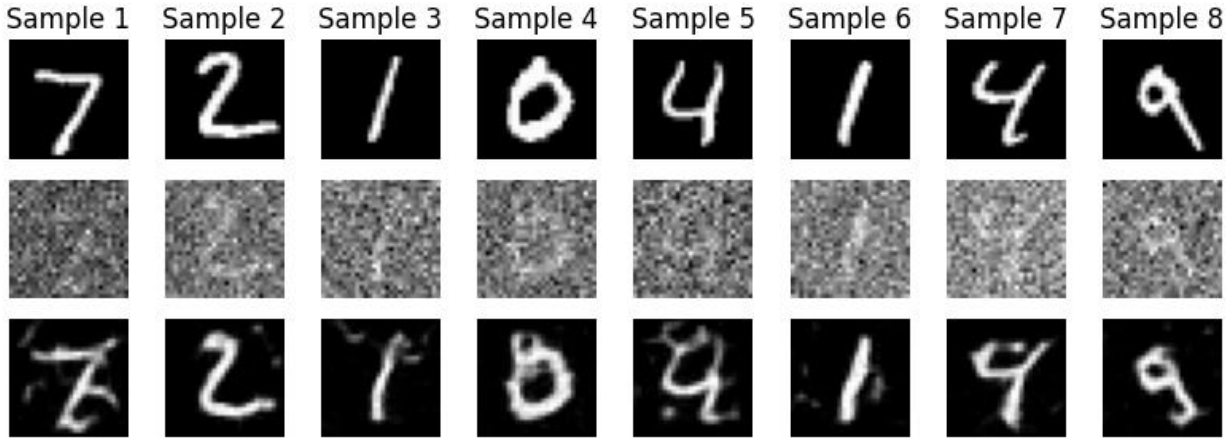
σ = 0.5



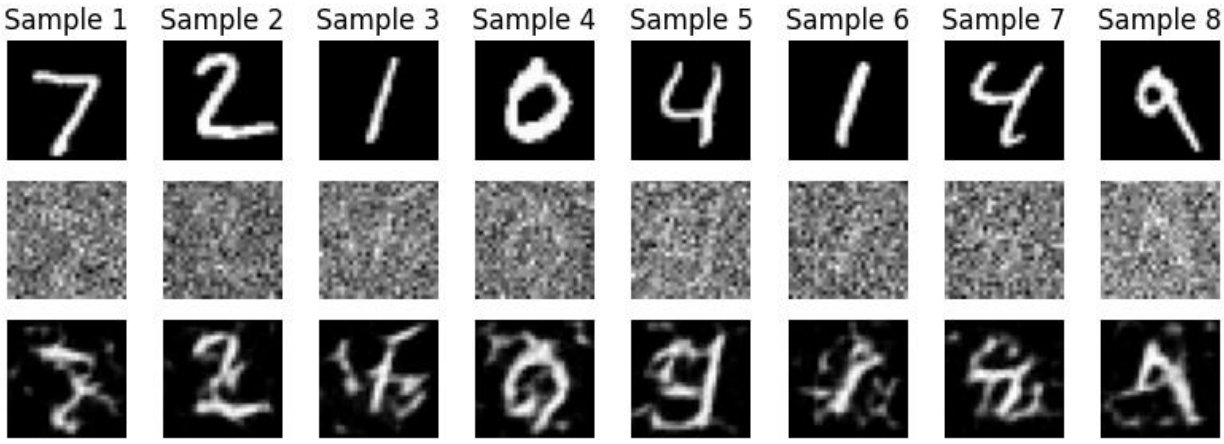
σ = 0.6



σ = 0.8

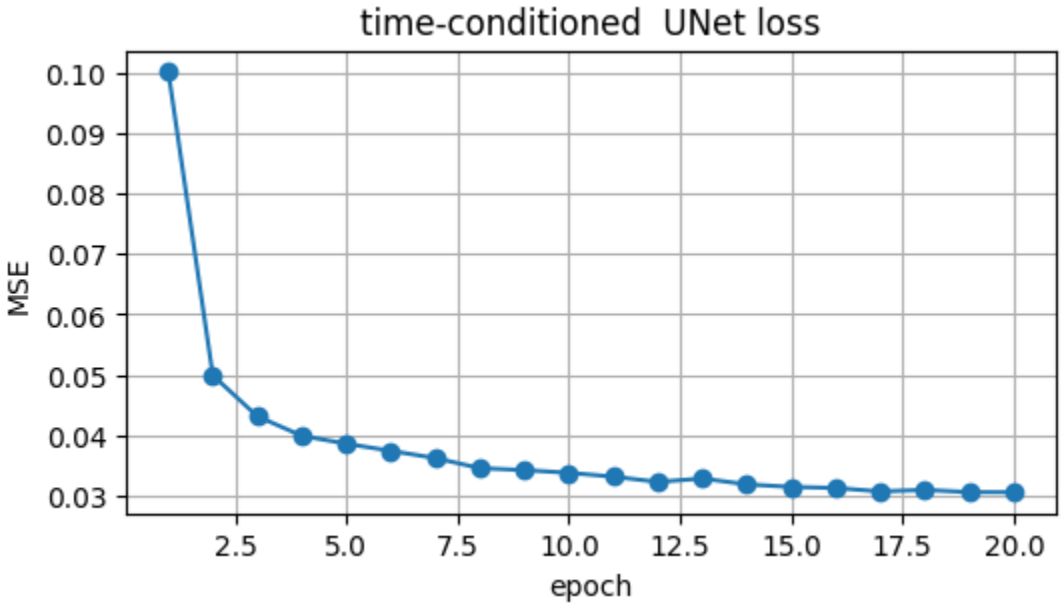


σ = 1.0

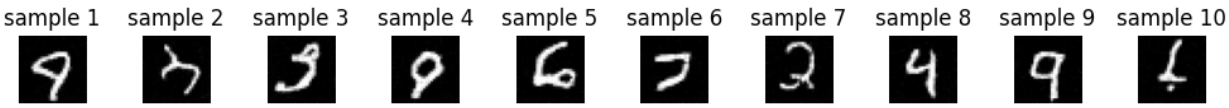

Sample 1
Sample 2
Sample 3
Sample 4
Sample 5
Sample 6
Sample 7
Sample 8

Part 2.1-2.3: Time-conditioned UNet

1. A training loss curve plot for the time-conditioned UNet over the whole training process (Figure 10).



2. Sampling results for the class-conditioned UNet after the 5th and the 20th epoch. Providing a GIF is optional and can be done as extra credit.
   5th:

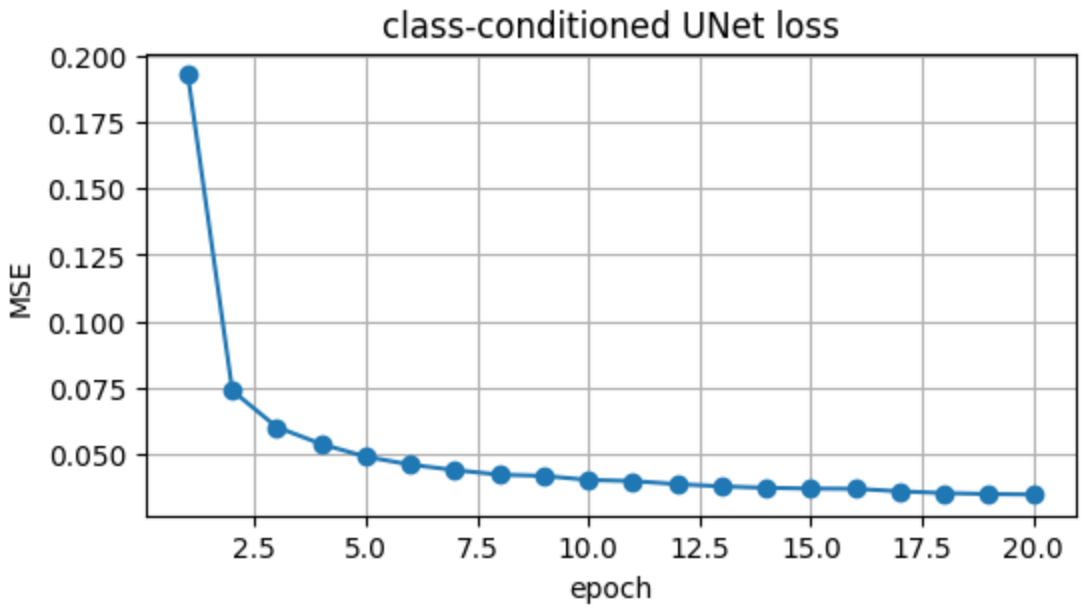


   20th:



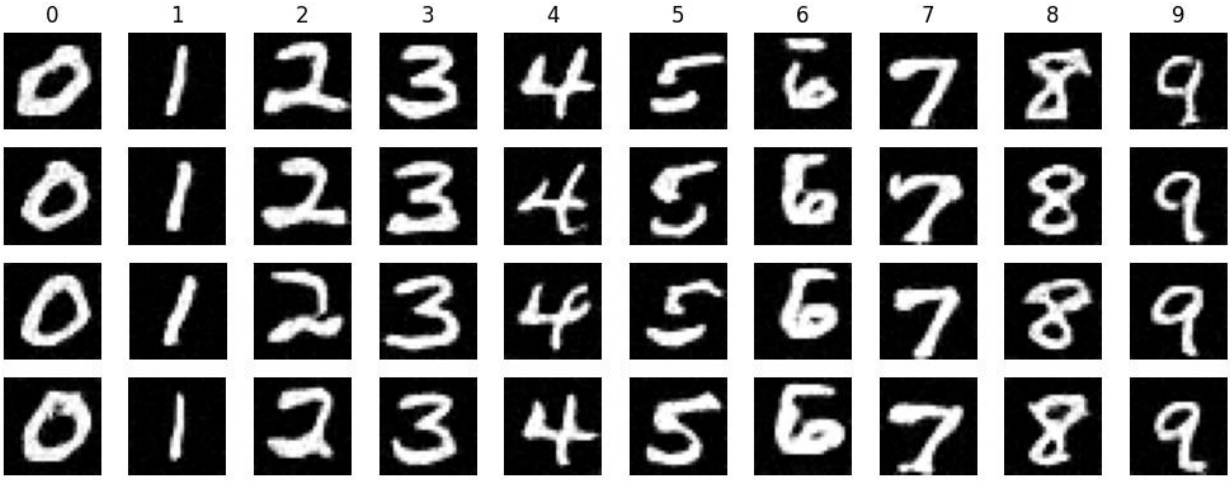
Part 2.4-2.5: Class-conditioned UNet

1. A training loss curve plot for the class-conditioned UNet over the whole training process.

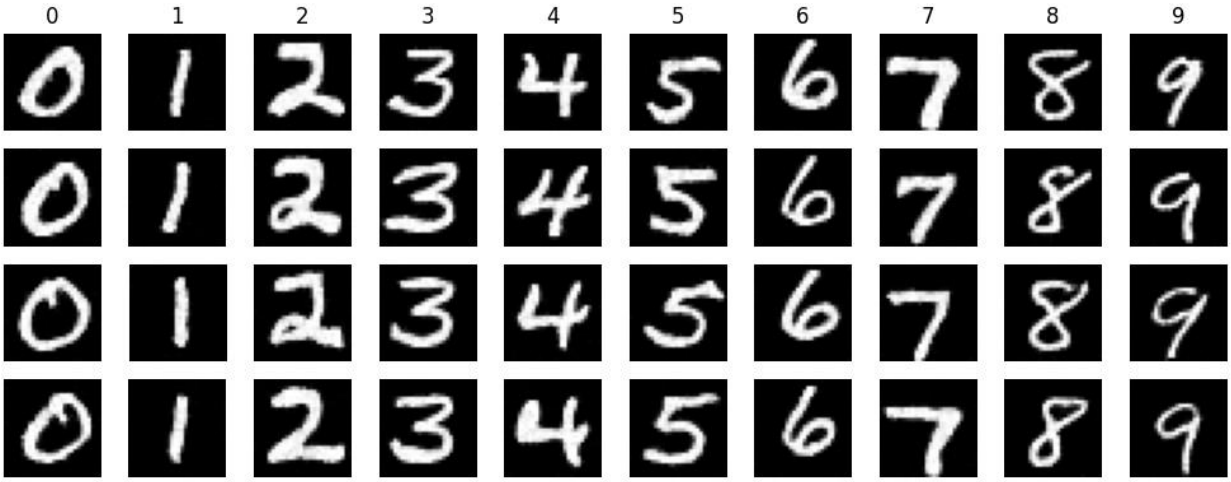


2. Sampling results for the class-conditioned UNet after the 5th and the 20th epoch. Generate 4 instances of each digit as shown on the assignment page. Providing a GIF is optional and can be done as extra credit.

   5th:



   20th:



Extra credit: